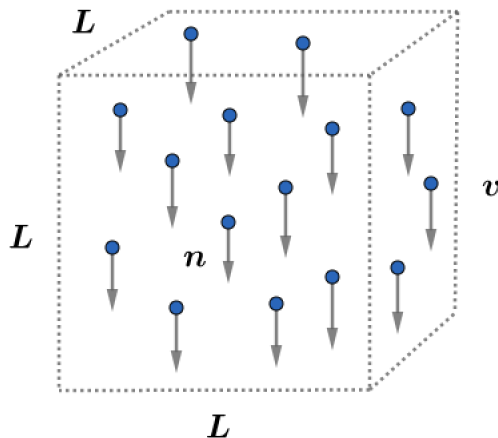


2018A F=ma Exam: Problem 19

Kevin S. Huang



Consider a simple model of a cube of raindrops hitting the ground. From Newton's 2nd law, we have

$$F = \frac{\Delta p}{\Delta t}$$

$$P = \frac{\Delta p}{A \Delta t}$$

Assuming the raindrops travel with constant velocity v , the time it takes the cube to collide into the ground is given by

$$\Delta t = \frac{L}{v}$$

The raindrops are all brought to rest so

$$\Delta p = m_{tot} v = n L^3 m v = n L^3 \left(\frac{4}{3} \pi r^3 \rho \right) v$$

Thus,

$$P = \frac{n L^3 \left(\frac{4}{3} \pi r^3 \rho \right) v}{L^2 \left(\frac{L}{v} \right)} = \frac{4}{3} \pi \rho n r^3 v^2$$

$$P' = \frac{4}{3} \pi \rho (2n) \left(\frac{r}{2} \right)^3 \left(\frac{v}{2} \right)^2 = \frac{P}{16}$$

so the answer is E.